

Supplementary material: HCA-DSE

Full budget sweeps and validation subsets accompanying the main manuscript. All values are generated from the same result CSVs as the main-text summaries.

Table S1: Full budget-matched comparison on S2. Medians over 20 seeds.

scale	workload	method	evals	recall	HV ratio	IGD ⁺	time [s]
S2	telemetry	exhaustive	3 888	1.000	1.000	0.0000	0.044
		random ($B=100$)	100	0.037	0.642	0.1455	0.001
		random ($B=250$)	250	0.074	0.795	0.1033	0.003
		random ($B=500$)	500	0.148	0.882	0.0602	0.006
		random ($B=1000$)	1000	0.259	0.927	0.0388	0.014
		NSGA-II ($B=100$)	100	0.018	0.580	0.1773	0.010
		NSGA-II ($B=250$)	250	0.241	0.883	0.0467	0.028
		NSGA-II ($B=500$)	500	0.685	0.981	0.0081	0.064
		NSGA-II ($B=1000$)	1000	0.926	0.999	0.0002	0.147
		HCA-DSE	44	1.000	1.000	0.0000	0.036
S2	control	exhaustive	3 888	1.000	1.000	0.0000	0.042
		random ($B=100$)	100	0.040	0.652	0.1598	0.001
		random ($B=250$)	250	0.080	0.791	0.1056	0.003
		random ($B=500$)	500	0.120	0.875	0.0556	0.006
		random ($B=1000$)	1000	0.240	0.922	0.0377	0.014
		NSGA-II ($B=100$)	100	0.020	0.594	0.1519	0.011
		NSGA-II ($B=250$)	250	0.320	0.902	0.0373	0.030
		NSGA-II ($B=500$)	500	0.720	0.993	0.0065	0.070
		NSGA-II ($B=1000$)	1000	0.800	0.999	0.0030	0.160
		HCA-DSE	37	1.000	1.000	0.0000	0.035
S2	sensing	exhaustive	3 888	1.000	1.000	0.0000	0.029
		random ($B=100$)	100	0.031	0.359	0.2624	0.001
		random ($B=250$)	250	0.062	0.750	0.1344	0.002
		random ($B=500$)	500	0.156	0.849	0.0859	0.005
		random ($B=1000$)	1000	0.250	0.897	0.0626	0.010
		NSGA-II ($B=100$)	100	0.000	0.271	0.3183	0.010
		NSGA-II ($B=250$)	250	0.250	0.725	0.0983	0.030
		NSGA-II ($B=500$)	500	0.625	0.941	0.0172	0.073
		NSGA-II ($B=1000$)	1000	0.875	1.000	0.0017	0.182
		HCA-DSE	26	1.000	1.000	0.0000	0.016

Table S2: Full budget-matched comparison on S3. Medians over 20 seeds.

scale	workload	method	evals	recall	HV ratio	IGD ⁺	time [s]
S3	telemetry	exhaustive	34 560	1.000	1.000	0.0000	0.308
		random ($B=100$)	100	0.000	0.372	0.3783	0.001
		random ($B=250$)	250	0.000	0.546	0.2558	0.002
		random ($B=500$)	500	0.000	0.613	0.1827	0.005
		random ($B=1000$)	1000	0.032	0.732	0.1147	0.011
		NSGA-II ($B=100$)	100	0.000	0.072	0.4807	0.010
		NSGA-II ($B=250$)	250	0.032	0.324	0.2723	0.026
		NSGA-II ($B=500$)	500	0.258	0.763	0.1132	0.061
		NSGA-II ($B=1000$)	1000	0.500	0.946	0.0454	0.145
		HCA-DSE	50	1.000	1.000	0.0000	0.138
S3	control	exhaustive	34 560	1.000	1.000	0.0000	0.265
		random ($B=100$)	100	0.000	0.365	0.4042	0.001
		random ($B=250$)	250	0.000	0.525	0.2933	0.002
		random ($B=500$)	500	0.000	0.609	0.1771	0.005
		random ($B=1000$)	1000	0.000	0.725	0.1189	0.010
		NSGA-II ($B=100$)	100	0.000	0.114	0.4329	0.010
		NSGA-II ($B=250$)	250	0.037	0.575	0.2181	0.027
		NSGA-II ($B=500$)	500	0.352	0.888	0.0427	0.064
		NSGA-II ($B=1000$)	1000	0.648	0.984	0.0236	0.147
		HCA-DSE	41	1.000	1.000	0.0000	0.125
S3	sensing	exhaustive	34 560	1.000	1.000	0.0000	0.266
		random ($B=100$)	100	0.000	0.045	0.5753	0.001
		random ($B=250$)	250	0.000	0.311	0.3251	0.002
		random ($B=500$)	500	0.000	0.446	0.2306	0.005
		random ($B=1000$)	1000	0.000	0.658	0.1569	0.010
		NSGA-II ($B=100$)	100	0.000	0.030	0.6768	0.010
		NSGA-II ($B=250$)	250	0.018	0.126	0.3325	0.027
		NSGA-II ($B=500$)	500	0.204	0.713	0.2267	0.064
		NSGA-II ($B=1000$)	1000	0.741	0.951	0.1677	0.175
		HCA-DSE	43	1.000	1.000	0.0000	0.085

Table S3: Full S3 statistical comparison: one-sided Mann–Whitney test for larger NSGA-II hypervolume and Cliff’s δ . Unadjusted p -values.

workload	budget	NSGA-II HV med.	random HV med.	p	Cliff’s δ	NSGA-II recall	HCA-DSE evals
telemetry	100	0.0721	0.3723	0.999	-0.565	0.0	50
telemetry	250	0.3235	0.546	0.938	-0.282	0.032	50
telemetry	500	0.7628	0.6125	0.338	0.08	0.258	50
telemetry	1000	0.9455	0.7322	0.299	0.1	0.5	50
control	100	0.1144	0.3648	0.975	-0.36	0.0	41
control	250	0.5746	0.5252	0.430	0.035	0.037	41
control	500	0.8884	0.6087	0.090	0.25	0.352	41
control	1000	0.9835	0.7246	0.057	0.295	0.648	41
sensing	100	0.0299	0.0449	0.955	-0.31	0.0	43
sensing	250	0.1262	0.3109	0.780	-0.14	0.018	43
sensing	500	0.7127	0.4465	0.308	0.095	0.204	43
sensing	1000	0.9506	0.6581	0.142	0.2	0.741	43

Table S4: Testbed validation against measured p_{50} , including utilisation and reproducibility subsets. Pair agreement is a fraction; undefined values (no eligible pairs) are shown as $-$.

subset	n	MAPE [%]	med. AE [ms]	Spearman ρ	Kendall τ	pair > 10%	pair > 25%	repeat IQR [ms]
telemetry (all)	12	19.9	7.43	0.811	0.697	0.815	0.895	0.23
telemetry ($\rho \leq 0.75$)	6	15.4	2.85	0.829	0.733	0.818	1.0	0.2
control (all)	12	14.4	2.49	0.865	0.748	0.896	0.909	0.11
control ($\rho \leq 0.75$)	10	9.4	2.32	0.766	0.629	0.815	0.833	0.11
sensing (all)	12	51.1	19.15	0.935	0.84	0.935	0.913	1.0
sensing ($\rho \leq 0.75$)	10	51.0	18.82	0.985	0.944	1.0	1.0	0.75
sensing (reproducible)	8	39.3	18.56	0.97	0.909	1.0	-	0.67
all workloads (all)	36	28.4	7.43	0.968	0.884	0.961	0.98	0.27
all workloads ($\rho \leq 0.75$)	26	26.8	3.96	0.945	0.855	0.959	0.982	0.22
all workloads (reproducible)	32	22.7	4.5	0.957	0.869	0.956	0.979	0.23

Table S5: Full order/queue ablation on S1–S3. All configurations preserve the unrounded exhaustive front.

scale	workload	order	queue	states	checks	evals	active	pruned	ms
S1	telemetry	O1	off	695	1299	136	0	0	9.7
S1	telemetry	O1	on	589	1299	30	0	0	10.6
S1	telemetry	O2	off	447	746	136	0	0	5.8
S1	telemetry	O2	on	249	516	32	306	46	4.8
S1	control	O1	off	679	1317	112	0	0	9.5
S1	control	O1	on	593	1317	26	0	0	10.5
S1	control	O2	off	391	669	113	0	0	5.3
S1	control	O2	on	242	512	28	312	35	4.7
S1	sensing	O1	off	497	819	148	0	0	5.8
S1	sensing	O1	on	365	819	16	0	0	5.7
S1	sensing	O2	off	481	810	147	0	0	5.9
S1	sensing	O2	on	226	503	16	309	72	3.7
S2	telemetry	O1	off	1816	3940	208	0	0	30.6
S2	telemetry	O1	on	1652	3940	44	0	0	33.4
S2	telemetry	O2	off	939	1815	208	0	0	15.1
S2	telemetry	O2	on	604	1383	46	839	99	13.0
S2	control	O1	off	1780	3960	164	0	0	29.9
S2	control	O1	on	1653	3960	37	0	0	34.2
S2	control	O2	off	835	1674	165	0	0	13.1
S2	control	O2	on	587	1364	39	842	80	12.0
S2	sensing	O1	off	1268	2421	290	0	0	17.5
S2	sensing	O1	on	1004	2421	26	0	0	15.8
S2	sensing	O2	off	1161	2184	289	0	0	15.1
S2	sensing	O2	on	600	1427	26	879	177	9.9
S3	telemetry	O1	off	6029	14794	251	0	0	113.0
S3	telemetry	O1	on	5828	14794	50	0	0	132.5
S3	telemetry	O2	off	2871	6806	251	0	0	51.2
S3	telemetry	O2	on	2397	6100	52	3777	188	53.4
S3	control	O1	off	5990	14812	205	0	0	108.6
S3	control	O1	on	5826	14812	41	0	0	129.0
S3	control	O2	off	2755	6636	206	0	0	48.2
S3	control	O2	on	2375	6074	43	3777	163	50.9
S3	sensing	O1	off	4888	11367	538	0	0	83.9
S3	sensing	O1	on	4393	11367	43	0	0	83.2
S3	sensing	O2	off	4351	9905	537	0	0	72.5
S3	sensing	O2	on	3221	8281	43	5139	419	60.8